

Quivira Math

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1 Establishing Euler-Lagrange equations formally

Let us consider a configuration space defined by q_h , $h = 1, \dots, N$, also captured in vector form by $\mathbf{q}$. Introducing the Lagrangian:

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}})$$

and the action:

$$S \triangleq \int_0^T \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) dt$$

we want to use calculus of variations to find the $\mathbf{q}(t)$ that result in a minimal action. In essence we want:

$$\min_{\mathbf{q} \in \mathcal{Q}} S \triangleq \min_{\mathbf{q} \in \mathcal{Q}} \int_0^T \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) dt$$

where $\mathcal{Q}$ is the functional space containing all possible trajectories, with fixed endpoints. If $\bar{\mathbf{q}}(t)$ denotes a stationary trajectory, a perturbed trajectory is written as $\mathbf{q}(t) = \bar{\mathbf{q}}(t) + \delta\mathbf{q}(t)$, where the variations satisfy the boundary conditions $\delta\mathbf{q}(0) = \delta\mathbf{q}(T) = \mathbf{0}$.

The first variation of the Lagrangian $\delta\mathcal{L}$, evaluated along $\bar{\mathbf{q}}(t)$, is, to first order in $\delta\mathbf{q}$:

$$\delta\mathcal{L} = \left. \frac{d}{d\varepsilon} \mathcal{L}(\bar{\mathbf{q}} + \varepsilon\delta\mathbf{q}, \dot{\bar{\mathbf{q}}} + \varepsilon\delta\dot{\mathbf{q}}) \right|_{\varepsilon=0} = \frac{\partial\mathcal{L}}{\partial\mathbf{q}}\delta\mathbf{q} + \frac{\partial\mathcal{L}}{\partial\dot{\mathbf{q}}}\delta\dot{\mathbf{q}}.$$

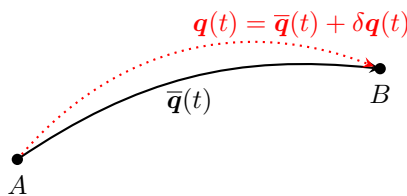


Figure 1: Nominal trajectory $\bar{\mathbf{q}}(t)$ (solid) and perturbed trajectory $\mathbf{q}(t) = \bar{\mathbf{q}}(t) + \delta\mathbf{q}(t)$ (dotted, red) from A to B . Note that $\delta\mathbf{q}(0) = \delta\mathbf{q}(T) = \mathbf{0}$.

Here and below, the partial derivatives of $\mathcal{L}$ are evaluated at $(\bar{\mathbf{q}}, \dot{\bar{\mathbf{q}}})$. We now require the first variation of the action δS to vanish:

$$\begin{aligned}
\delta S &= \int_0^T \left(\frac{\partial \mathcal{L}}{\partial \mathbf{q}} \delta \mathbf{q} + \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \delta \dot{\mathbf{q}} \right) dt \\
&= \int_0^T \frac{\partial \mathcal{L}}{\partial \mathbf{q}} \delta \mathbf{q} dt + \int_0^T \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \frac{d}{dt} (\delta \mathbf{q}) dt \\
&= \int_0^T \frac{\partial \mathcal{L}}{\partial \mathbf{q}} \delta \mathbf{q} dt + \left. \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \delta \mathbf{q} \right|_0^T - \int_0^T \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \right) \delta \mathbf{q} dt \\
&= \int_0^T \left[\frac{\partial \mathcal{L}}{\partial \mathbf{q}} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \right) \right] \delta \mathbf{q} dt = 0.
\end{aligned}$$

The boundary term vanishes because $\delta \mathbf{q}(0) = \delta \mathbf{q}(T) = \mathbf{0}$. Since $\delta \mathbf{q}$ is otherwise arbitrary, the Euler–Lagrange equations follow:

$$\boxed{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{0}.}$$

2 Adding holonomic constraints

Let us now assume that the configuration space introduced above, of dimension N , is restricted by $M < N$ holonomic, time-independent constraints:

$$\mathbf{F}(\mathbf{q}) = \mathbf{0}$$

where $\mathbf{F}(\mathbf{q}) \in \mathbb{R}^M$. We can account for these constraints by introducing a vector of time-dependent Lagrange multipliers $\boldsymbol{\lambda}(t) \in \mathbb{R}^M$ and defining the constrained action:

$$\tilde{S} = \int_0^T [\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{\lambda}(t) \cdot \mathbf{F}(\mathbf{q})] dt.$$

The multipliers must be functions of time because the constraints have to be satisfied at every instant along the trajectory.

Following the same variational argument as in the previous section, the first variation of the constrained action is:

$$\delta \tilde{S} = \int_0^T [\delta \mathcal{L} + \delta \boldsymbol{\lambda} \cdot \mathbf{F}(\mathbf{q}) + \boldsymbol{\lambda} \cdot \delta \mathbf{F}(\mathbf{q})] dt.$$

Since the constraints depend only on $\mathbf{q}$, their variation is:

$$\delta \mathbf{F}(\mathbf{q}) = \frac{\partial \mathbf{F}}{\partial \mathbf{q}} \delta \mathbf{q}.$$

Therefore:

$$\delta \tilde{S} = \int_0^T \left[\frac{\partial \mathcal{L}}{\partial \mathbf{q}} \delta \mathbf{q} + \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \delta \dot{\mathbf{q}} + \boldsymbol{\lambda} \cdot \frac{\partial \mathbf{F}}{\partial \mathbf{q}} \delta \mathbf{q} + \delta \boldsymbol{\lambda} \cdot \mathbf{F}(\mathbf{q}) \right] dt.$$

Integrating by parts the term containing $\delta\dot{\mathbf{q}}$, and using the boundary conditions $\delta\mathbf{q}(0) = \delta\mathbf{q}(T) = \mathbf{0}$, gives:

$$\begin{aligned}\delta\tilde{S} = & \int_0^T \left[\frac{\partial\mathcal{L}}{\partial\mathbf{q}} - \frac{d}{dt} \left(\frac{\partial\mathcal{L}}{\partial\dot{\mathbf{q}}} \right) + \boldsymbol{\lambda} \cdot \frac{\partial\mathbf{F}}{\partial\mathbf{q}} \right] \delta\mathbf{q} dt \\ & + \int_0^T \delta\boldsymbol{\lambda} \cdot \mathbf{F}(\mathbf{q}) dt = 0.\end{aligned}$$

The variations $\delta\mathbf{q}$ and $\delta\boldsymbol{\lambda}$ are independent and arbitrary. Consequently, variation with respect to $\delta\boldsymbol{\lambda}$ gives back the constraint equations:

$$\mathbf{F}(\mathbf{q}) = \mathbf{0}.$$

Using the numerator-layout convention (heyoka compatible), $\mathbf{J} \triangleq \frac{\partial\mathbf{F}}{\partial\mathbf{q}}$ is the $M \times N$ constraint Jacobian. Therefore, the generalized constraint force is written as:

$$\left(\frac{\partial\mathbf{F}}{\partial\mathbf{q}} \right)^T \boldsymbol{\lambda} = \mathbf{J}(\mathbf{q})^T \boldsymbol{\lambda}$$

where

$$\mathbf{J}(\mathbf{q}) \triangleq \frac{\partial\mathbf{F}}{\partial\mathbf{q}}.$$

Considering simultaneously variations with respect to $\delta\mathbf{q}$ and $\delta\boldsymbol{\lambda}$ then returns the constrained Euler-Lagrange equations:

$$\boxed{\begin{cases} \frac{d}{dt} \left(\frac{\partial\mathcal{L}}{\partial\dot{\mathbf{q}}} \right) - \frac{\partial\mathcal{L}}{\partial\mathbf{q}} = \mathbf{J}(\mathbf{q})^T \boldsymbol{\lambda} \\ \mathbf{F}(\mathbf{q}) = \mathbf{0} \end{cases}} \quad (1)$$

The constrained dynamics of holonomic systems are therefore described by a differential-algebraic equation system.

3 Computing $\boldsymbol{\lambda}$

Introduce the mass matrix:

$$\mathbf{M}(\mathbf{q}) \rightarrow M_{ij} \triangleq \frac{\partial^2\mathcal{L}}{\partial\dot{q}_i\partial\dot{q}_j}$$

which we assume, not essentially, to be only dependent on the configuration, not on the velocities, and

$$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \rightarrow C_{ij} \triangleq \frac{\partial^2\mathcal{L}}{\partial\dot{q}_i\partial q_j}$$

as well as the column vector:

$$\mathbf{B}(\mathbf{q}, \dot{\mathbf{q}}) \rightarrow B_i = \frac{\partial\mathcal{L}}{\partial q_i}$$

then the Euler-Lagrange equations become:

$$\begin{cases} M\ddot{\mathbf{q}} + C\dot{\mathbf{q}} - \mathbf{B} = \mathbf{J}^T \boldsymbol{\lambda} \\ \mathbf{F}(\mathbf{q}) = \mathbf{0} \end{cases}$$

Deriving the holonomic constraint twice we have:

$$\mathbf{J}\dot{\mathbf{q}} = \mathbf{0}$$

and

$$\dot{\mathbf{J}}\dot{\mathbf{q}} + \mathbf{J}\ddot{\mathbf{q}} = \mathbf{0}$$

where:

$$\dot{J}_{ij} = \frac{\partial F_i}{\partial q_j} \dot{q}_k$$

we can thus reduce the differential algebraic system eliminating $\boldsymbol{\lambda}$ from the saddle system:

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^T \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} -C\dot{\mathbf{q}} + \mathbf{B} \\ -\dot{\mathbf{J}}\dot{\mathbf{q}} \end{bmatrix}$$

Using a Runge-Kutta type of integration scheme one could solve the saddle system numerically at each step and be finished.

4 An expression for the rhs

Let

$$\mathbf{f} \triangleq \mathbf{B} - C\dot{\mathbf{q}}, \quad \ddot{\mathbf{q}}_{\text{free}} \triangleq \mathbf{M}^{-1} \mathbf{f}, \quad \mathbf{W} \triangleq \mathbf{J} \mathbf{M}^{-1} \mathbf{J}^T. \quad (2)$$

where $\mathbf{W}$ is called the Delassus matrix and can be interpreted as the constraint-space inverse inertia matrix. In particular, it maps the vector of Lagrange multipliers, which represent constraint reactions, to the acceleration induced along the constraint directions. The first block row of the saddle-point system gives

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_{\text{free}} + \mathbf{M}^{-1} \mathbf{J}^T \boldsymbol{\lambda}.$$

Using the differentiable version of the constraints:

$$\mathbf{J}\ddot{\mathbf{q}} = -\dot{\mathbf{J}}\dot{\mathbf{q}},$$

the Lagrange multipliers are obtained as

$$\boldsymbol{\lambda} = -\mathbf{W}^{-1} \left(\mathbf{J}\ddot{\mathbf{q}}_{\text{free}} + \dot{\mathbf{J}}\dot{\mathbf{q}} \right). \quad (3)$$

Hence, the constrained acceleration is

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_{\text{free}} - \mathbf{M}^{-1} \mathbf{J}^T \mathbf{W}^{-1} \left(\mathbf{J}\ddot{\mathbf{q}}_{\text{free}} + \dot{\mathbf{J}}\dot{\mathbf{q}} \right). \quad (4)$$

5 Adding nonconservative forces

The preceding derivation assumes that all applied forces are conservative and therefore represented by the Lagrangian

$$\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) = T(\mathbf{q}, \dot{\mathbf{q}}) - V(\mathbf{q}).$$

In practice, a system may also be subject to forces and torques which cannot, or should not, be represented through a scalar potential. Examples include viscous damping, dry friction, actuator forces, thruster forces, aerodynamic loads, prescribed motor torques, and control inputs.

Such loads enter the equations of motion through the Lagrange-d'Alembert principle. Let

$$\mathbf{Q}_{\text{nc}}(\mathbf{q}, \dot{\mathbf{q}}, t) \in \mathbb{R}^N$$

denote the vector of generalized nonconservative forces. The virtual work of all nonconservative loads must be expressible as

$$\boxed{\delta W_{\text{nc}} = \mathbf{Q}_{\text{nc}}(\mathbf{q}, \dot{\mathbf{q}}, t)^\top \delta \mathbf{q}.} \quad (5)$$

Equation (5) is the defining requirement for the generalized load vector. In particular, its i th component is the force which is virtual-work conjugate to the generalized coordinate q_i :

$$\delta W_{\text{nc}} = \sum_{i=1}^N Q_{\text{nc},i} \delta q_i.$$

Thus, if q_i is translational, $Q_{\text{nc},i}$ has units of force; if q_i is an angle, $Q_{\text{nc},i}$ has units of torque.

The Lagrange-d'Alembert principle for the constrained system is

$$\delta \tilde{S} + \int_0^T \delta W_{\text{nc}} dt = 0, \quad (6)$$

where

$$\tilde{S} = \int_0^T [\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}) + \boldsymbol{\lambda}^\top \mathbf{F}(\mathbf{q})] dt.$$

Following the same integration-by-parts argument as before gives

$$\boxed{\begin{cases} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{Q}_{\text{nc}} + \mathbf{J}(\mathbf{q})^\top \boldsymbol{\lambda} \\ \mathbf{F}(\mathbf{q}) = \mathbf{0}. \end{cases}} \quad (7)$$

The nonconservative generalized forces are therefore added directly to the right-hand side of the Euler-Lagrange equations, whereas $\mathbf{J}^\top \boldsymbol{\lambda}$ remains the generalized constraint reaction.

Using the definitions introduced previously,

$$M\ddot{\mathbf{q}} + C\dot{\mathbf{q}} - \mathbf{B} = \mathbf{Q}_{\text{nc}} + \mathbf{J}^\top \boldsymbol{\lambda},$$

or equivalently,

$$M\ddot{\mathbf{q}} = \mathbf{f}_{\text{nc}} + \mathbf{J}^\top \boldsymbol{\lambda}, \quad \mathbf{f}_{\text{nc}} \triangleq \mathbf{B} - C\dot{\mathbf{q}} + \mathbf{Q}_{\text{nc}}. \quad (8)$$

The saddle-point system becomes

$$\boxed{\begin{bmatrix} \mathbf{M} & -\mathbf{J}^\top \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{B} - C\dot{\mathbf{q}} + \mathbf{Q}_{\text{nc}} \\ -\mathbf{J}\dot{\mathbf{q}} \end{bmatrix}}. \quad (9)$$

Accordingly, the free acceleration—that is, the acceleration in the absence of constraint reactions but including nonconservative applied loads—is

$$\ddot{\mathbf{q}}_{\text{free}} \triangleq \mathbf{M}^{-1} \mathbf{f}_{\text{nc}} = \mathbf{M}^{-1} (\mathbf{B} - C\dot{\mathbf{q}} + \mathbf{Q}_{\text{nc}}). \quad (10)$$

The expressions for the Lagrange multipliers and the constrained acceleration retain exactly the same form

5.1 Generalized forces induced by physical forces and torques

A user may directly provide

$$\mathbf{Q}_{\text{nc}}(\mathbf{q}, \dot{\mathbf{q}}, t),$$

provided that it satisfies the virtual-work relation in (5). This is the most general and minimal interface. However, physical loads are usually specified as forces acting at points and torques acting on rigid bodies. Let a force $\mathbf{f}_P$ be applied at a point P , whose inertial position is

$$\mathbf{r}_P = \mathbf{r}_P(\mathbf{q}).$$

Its virtual displacement is

$$\delta \mathbf{r}_P = \frac{\partial \mathbf{r}_P}{\partial \mathbf{q}} \delta \mathbf{q}.$$

Define the point-position Jacobian

$$\mathbf{J}_{r,P}(\mathbf{q}) \triangleq \frac{\partial \mathbf{r}_P}{\partial \mathbf{q}} \in \mathbb{R}^{3 \times N}. \quad (11)$$

The virtual work done by the applied force is

$$\delta W_{\mathbf{f}_P} = \mathbf{f}_P^\top \delta \mathbf{r}_P = \mathbf{f}_P^\top \mathbf{J}_{r,P} \delta \mathbf{q} = (\mathbf{J}_{r,P}^\top \mathbf{f}_P)^\top \delta \mathbf{q}. \quad (12)$$

Therefore, the generalized force induced by a physical force applied at P is

$$\boxed{\mathbf{Q}_{\mathbf{f}_P} = \mathbf{J}_{r,P}^\top \mathbf{f}_P}. \quad (13)$$

Similarly, let a pure torque, or free couple, $\boldsymbol{\tau}_B$ be applied to a rigid body B . Let $\boldsymbol{\omega}_B$ denote the angular velocity of the body, and define the angular-velocity Jacobian by

$$\boldsymbol{\omega}_B = \mathbf{J}_{\omega,B}(\mathbf{q})\dot{\mathbf{q}}, \quad \mathbf{J}_{\omega,B}(\mathbf{q}) \in \mathbb{R}^{3 \times N}. \quad (14)$$

The corresponding virtual rotation is

$$\delta\boldsymbol{\theta}_B = \mathbf{J}_{\omega,B}\delta\mathbf{q}.$$

The virtual work of the applied torque is consequently

$$\delta W_{\boldsymbol{\tau}_B} = \boldsymbol{\tau}_B^\top \delta\boldsymbol{\theta}_B = \boldsymbol{\tau}_B^\top \mathbf{J}_{\omega,B} \delta\mathbf{q} = (\mathbf{J}_{\omega,B}^\top \boldsymbol{\tau}_B)^\top \delta\mathbf{q}. \quad (15)$$

The induced generalized force is therefore

$$\boxed{\mathbf{Q}_{\boldsymbol{\tau}_B} = \mathbf{J}_{\omega,B}^\top \boldsymbol{\tau}_B.} \quad (16)$$